Hereditary Sensory and Autonomic Neuropathy Type VI (HSAN-VI): Comprehensive Disease Characterization Report

Disease: Hereditary Sensory and Autonomic Neuropathy Type 6 (HSAN-VI) **MONDO ID:** MONDO:0013839 | **OMIM:** #614653 | **Orphanet:** ORPHA:314381 **Causal gene:** *DST* (dystonin / BPAG1) | **Inheritance:** Autosomal recessive **Category:** Mendelian (ultra-rare)

Summary

Hereditary Sensory and Autonomic Neuropathy Type VI (HSAN-VI) is an ultra-rare, autosomal recessive disorder of the peripheral nervous system caused by biallelic loss-of-function of the **neuronal DST-a isoforms** of the *DST* gene (dystonin, also known as bullous pemphigoid antigen 1 / BPAG1). Dystonin-a proteins are large plakin-family **cytoskeletal cross-linkers (spectraplakins)** that physically tether the actin microfilament, intermediate filament (neurofilament), and microtubule networks within sensory and autonomic neurons. When these isoforms are lost, the axonal cytoskeleton disorganizes, axonal transport fails, the endoplasmic reticulum/nuclear envelope becomes dysfunctional, and autophagic flux is impaired — collectively driving progressive degeneration of dorsal root ganglion (DRG) sensory neurons and autonomic neurons. As of 2025, the disease remains vanishingly rare, with only about 15 reported patients carrying 11 distinct *DST* variants (mostly compound heterozygous).

Clinically, HSAN-VI presents on a **severity spectrum graded by which DST-a isoforms are ablated**. At the severe end lies fetal/neonatal lethal disease — arthrogryposis multiplex congenita (now formalized as lethal congenital contracture syndrome 12, LCCS12), severe neonatal hypotonia, respiratory and feeding failure, profound pain insensitivity, and prominent dysautonomia (alacrima, labile blood pressure and temperature, cardiac arrhythmia). At the milder end is adult-onset, slowly progressive sensory neuropathy with

painless ulcers. Residual expression of **DST-A3** acts as an endogenous protective modifier that preserves microtubule stability and softens the phenotype. *DST* is one member of a four-disorder single-gene allelic series, along with epidermolysis bullosa simplex 3 (EBS3, from loss of the epithelial DST-e isoform), congenital myopathy 29 (CMYO29, from loss of the muscular DST-b isoform), and LCCS12.

Diagnosis is molecular — exome/genome sequencing or hereditary-neuropathy gene panels, with copy-number analysis needed given the very large *DST* locus. There is **no approved disease-modifying therapy**; management is entirely supportive and multidisciplinary (airway/respiratory support, nutrition, corneal protection, wound/ulcer prevention, autonomic monitoring). The **dystonia musculorum (dt) mouse** faithfully recapitulates the human disease and has provided proof-of-concept that neuronal DST-a2 gene restoration partially rescues the phenotype, defining a rational future therapeutic direction.

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1. Disease Information

Overview. HSAN-VI is an autosomal recessive subtype within the hereditary sensory and autonomic neuropathy (HSAN) family. It manifests severe pain insensitivity, neonatal hypotonia, respiratory and feeding difficulties, lack of psychomotor development, and autonomic abnormalities. The underlying cause is pathogenic variants in *DST*, which encodes dystonin (PMID: 40938507):

> "HSAN-VI (OMIM: 614653) is an autosomal recessive subtype which manifests severe pain insensitivity, neonatal hypotonia, respiratory and feeding difficulties, lack of psychomotor development and autonomic abnormalities. Pathogenic variants in the DST gene, which encodes dystonin have been identified as the underlying cause of HSAN-VI. Thus far, only 15 HSAN-VI patients associated with 11 DST variants, mostly compound heterozygous, have been identified in the literature."

Key identifiers (database-verified).

Resource	Identifier
MONDO	MONDO:0013839
OMIM	#614653
Orphanet	ORPHA:314381
Disease Ontology	DOID:0070151
GARD	GARD:0012987
MedGen	761278
UMLS	C3539003
ICD-10	G60.8 (group level only)
ICD-11	8C10.Y (group level only)
MeSH	D009477 — *Hereditary Sensory and Autonomic Neuropathies* (group level only)

Importantly, **no ICD or MeSH code exists specifically for HSAN-VI** — those classifications apply only at the broader HSAN group level; the granular MONDO term carries no ICD/MeSH mapping.

Synonyms / alternative names. HSAN VI; HSAN type 6; familial dysautonomia with contractures; DST hereditary sensory and autonomic neuropathy.

Information source. The disease-level knowledge base entry is derived from **aggregated disease-level resources** (OMIM, Orphanet, MONDO, HPO/JAX) combined with a small number of **individual patient case reports** — reflecting the ultra-rare nature of the disorder.

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2. Etiology

Primary cause — genetic. HSAN-VI is a monogenic Mendelian disorder caused by **biallelic (recessive) loss-of-function variants in the neuronal DST-a isoforms** of *DST*. There is no known environmental, infectious, or acquired cause. Disease requires two damaged alleles; heterozygous carriers are asymptomatic.

Genetic risk factors. The causal variants are within *DST* itself. *DST* is strongly loss-of-function constrained at the gene level (gnomAD: LOEUF ≈ 0.32, observed/expected LoF = 0.29 [253 observed vs 875.2 expected], pLI = 1, missense Z = 3.68), consistent with an essential developmental gene. The specific set of DST-a isoforms disrupted by a given variant is the principal determinant of severity (see Sections 3, 4, 6).

Modifier genes / modifier alleles. The clearest modifier is **isoform-intrinsic**: residual expression of DST-A3 partially compensates for loss of DST-A1/A2 and reduces severity (PMID: 29982604). No independent trans-acting modifier gene has been established.

Environmental risk / protective factors. None identified — this is a fully penetrant Mendelian disorder. There are no reported environmental toxins, lifestyle factors, or protective exposures. **Consanguinity** raises the risk of homozygosity (several reported cases are from consanguineous unions).

Gene–environment interactions. Not applicable / none reported.

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3. Phenotypes

HSAN-VI is annotated by HPO/JAX with **39 phenotype terms** under OMIM:614653. Core features span sensory, autonomic, respiratory, gastrointestinal, and skeletal domains.

Domain	Phenotype	HPO term
Sensory	Sensory neuropathy / pain insensitivity	HP:0000763
Sensory	Areflexia	HP:0001284
Neurodevelopment	Profound global developmental delay	HP:0012736
Autonomic	Alacrima (reduced tears)	HP:0000522
Autonomic	Absent corneal reflex	HP:0034252
Autonomic (secondary)	Corneal scarring	HP:0000559
Autonomic	Hyperhidrosis	HP:0000975
Autonomic	Tachycardia	HP:0001649
Autonomic	Bradycardia	HP:0001662
Autonomic	Hypertension / labile BP	HP:0000822
Autonomic	Hyperpyrexia / temperature instability	HP:0033031
Respiratory	Neonatal respiratory distress	HP:0002643
Respiratory	Apnea	HP:0002104
Respiratory	Stridor	HP:0010307
Respiratory	Aspiration	HP:0002835
Respiratory	Bilateral vocal cord paresis	HP:0012822
Motor tone	Hypotonia (generalized/neonatal/axial)	HP:0001319 / HP:0001290 / HP:0001252
GI	Gastroesophageal reflux	HP:0002020
GI	Feeding difficulties	HP:0011968
GI	Poor suck	HP:0002033
Skeletal	Talipes equinovarus (club foot)	HP:0001762

Domain	Phenotype	HPO term
Skeletal	Flexion contracture	HP:0001371
Inheritance	Autosomal recessive inheritance	HP:0000007

Phenotype characteristics. - **Age of onset:** Predominantly **congenital/neonatal** in the severe form (arthrogryposis, hypotonia, respiratory distress at birth); the mild allelic form can be **adult-onset**. - **Severity: Variable** — from intrauterine/neonatal lethal to slowly progressive adult neuropathy. - **Progression: Progressive** neuronal degeneration; severe forms are rapidly fatal, milder forms slowly worsen. - **Frequency:** In the severe neonatal phenotype, pain insensitivity, hypotonia, respiratory/feeding difficulty, and autonomic instability are near-universal among reported cases.

Quality-of-life impact. Profound. In severe disease, affected infants have no meaningful psychomotor development, require ventilatory and nutritional support, and are at constant risk from autonomic crises, aspiration, and self-injury/corneal damage due to insensitivity to pain. The dysautonomia closely mirrors that of Riley–Day syndrome (familial dysautonomia).

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4. Genetic / Molecular Information

Causal gene. *DST* (dystonin / BPAG1). HGNC:1090; NCBI Gene 667; Ensembl ENSG00000151914; gene OMIM 113810; UniProt Q03001; located at **chr6p12.1** (GRCh38 chr6:56,457,981–56,955,274, minus strand). *DST* is very large and produces multiple tissue-specific isoform classes by alternative splicing: **DST-a (neuronal, a1/a2/a3)**, **DST-b (muscular, b1/b2/b3)**, and **DST-e (epithelial)** (PMID: 34897952, PMID: 37603210). HSAN-VI arises specifically from **loss of the neuronal DST-a isoforms** (PMID: 35276021):

> "This mutation resides within the plakin domain of BPAG1 and ablates all isoforms of this protein, leading to novel extracutaneous phenotypes consistent with HSAN-VI"

The DST four-disorder allelic series. A key organizing insight is that *DST* is a single locus underlying four distinct allelic disorders, resolvable by which isoform(s) a variant disrupts (PMID: 40497796):

Disorder	Isoform lost	Phenotype
HSAN-VI	DST-a (neuronal)	Sensory/autonomic neuropathy
EBS3 / EBSB2	DST-e (epithelial)	Epidermolysis bullosa simplex
CMYO29 (congenital myopathy 29)	DST-b (muscular)	Neonatal myopathy, arthrogryposis, dilated cardiomyopathy
LCCS12 (lethal congenital contracture syndrome 12)	DST-a + DST-b	Severe fetal/neonatal lethal arthrogryposis

PMID: 40497796: *"We propose redefining DST as a disease-associated gene linked to four distinct allelic disease phenotypes ... The location of the variant within DST allows for phenotype prediction."*

Pathogenic variants. Reported HSAN-VI variants are **compound heterozygous or homozygous LoF and missense** changes affecting DST-a. Representative pathogenic/likely-pathogenic examples (ClinVar) include: - c.22558del p.(Leu7520fs) — frameshift - p.His269Arg / c.905A>G p.(His302Arg) — missense - p.Arg1269Ter — nonsense (also annotated to LCCS12) - p.Ala203Glu — missense in the actin-binding domain; recombinant protein loses actin binding (PMID: 30371979) - c.1118C>T p.Pro373Leu — homozygous, consanguineous (PMID: 34897952)

Variant types: missense, frameshift, nonsense, and splice variants. **Classification** per ACMG/AMP: pathogenic and likely pathogenic records dominate ClinVar for HSAN-VI. **Allele frequency:** individual pathogenic alleles are extremely rare/private in gnomAD, consistent with the disorder's rarity and LoF constraint. **Origin:** germline. **Functional consequence: loss of function** (loss of the neuronal cytolinker).

Modifier genes / epigenetics / chromosomal abnormalities. The principal modifier is the **DST-A3 isoform** (intragenic compensation). No specific epigenetic mechanism (DNA methylation, histone modification) or recurrent large-scale chromosomal abnormality has been described for HSAN-VI. Given the size of *DST*, however, copy-number/structural analysis is diagnostically relevant.

5. Environmental Information

Not applicable. HSAN-VI is a purely genetic Mendelian disorder. There are **no established environmental factors, lifestyle factors, or infectious agents** that cause, trigger, or protect against the disease. Consanguinity is a demographic risk factor for recessive homozygosity but is not an environmental exposure per se.

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6. Mechanism / Pathophysiology

Ordered causal chain (mutation → clinical manifestation)

1. **Biallelic LoF variants in the neuronal DST-a isoforms** *lead to* absence or dysfunction of dystonin-a cytolinker protein in sensory and autonomic neurons.
2. Loss of dystonin-a *results in* **failure to physically tether actin microfilaments to intermediate filaments (neurofilaments) and microtubules** — the protein's core function (PMID: 8752219).
3. Cytolinker loss *leads to* **disorganization of the axonal cytoskeleton**, including disrupted actin organization (patient cells; p.Ala203Glu abolishes actin binding) and loss/aggregation of neurofilament proteins such as α-internexin (PMID: 30371979, PMID: 16691115).
4. Cytoskeletal disorganization *results in* **impaired fast axonal transport** in both orthograde and retrograde directions, with focal axonal swellings (PMID: 12859670).
5. In parallel, dystonin loss *results in* **endoplasmic reticulum / nuclear-envelope dysfunction and ER stress** (PMID: 18638474) **and impaired autophagic flux** (autophagosome accumulation, elevated LC3-II) (PMID: 26043942).
6. Loss of microtubule-stabilizing function *leads to* **microtubule instability with reduced tubulin acetylation** in severe alleles — partially rescued when DST-A3 is upregulated (PMID: 29982604).
7. These converging insults *drive* **progressive degeneration of DRG sensory neurons** (and autonomic neurons), with caspase-3 activation in aggregate-laden neurons (PMID: 16691115) — selectively affecting NT-3-responsive muscle afferents and GDNF-responsive neurons (PMID: 11241383).
8. **Sensory neuron loss** *manifests as* pain insensitivity, areflexia, and sensory ataxia; **muscle-spindle/afferent degeneration** *manifests as* hypotonia and ataxia (PMID: 9242412).

9. **Branch — autonomic:** degeneration of autonomic/afferent circuits *results in* dysautonomia — alacrima, labile blood pressure and temperature, and cardiac arrhythmia via disrupted afferent neural circuits (PMID: 42028226) — plus GI dysmotility (PMID: 31814231).

Detail

Molecular architecture (UniProt Q03001, canonical 7570 aa). Dystonin-a is a spectraplakin with an N-terminal tandem calponin-homology **actin-binding domain (CH1–CH2)**, an SH3 domain, 37 spectrin/plakin repeats, two EF-hands (calcium binding), and a C-terminal **GAR microtubule-binding domain**. This bipartite architecture is what allows it to bridge actin at one end and microtubules/intermediate filaments at the other (PMID: 8752219, PMID: 26778567).

GO annotations. - *Molecular function:* microtubule binding (GO:0008017), microtubule plus-end binding (GO:0051010), calcium ion binding (GO:0005509), integrin binding (GO:0005178). - *Biological process:* retrograde axonal transport (GO:0008090), cytoskeleton organization (GO:0007010). - *Cellular component:* axon (GO:0030424), ER membrane (GO:0005789), nuclear envelope (GO:0005635), microtubule/IF/actin cytoskeleton.

Cellular processes: cytoskeletal disorganization, impaired intracellular transport, ER stress, impaired autophagy, and ultimately apoptosis. **Upstream** events are the cytolinker loss and cytoskeletal failure; **downstream** events are transport failure, organelle stress, and neuronal death. **Cell types involved:** DRG sensory neurons (CL:0000101 sensory neuron), autonomic neurons, and Schwann cells (CL:0002573) — Schwann-cell-specific loss produces a late-onset neuropathy with hypomyelination (PMID: 32445516).

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7. Anatomical Structures Affected

Organ / system level. The **peripheral nervous system** is primary — specifically the **dorsal root ganglia** (UBERON:0000044) and the **sensory and autonomic divisions** of the PNS. Secondary/complication involvement spans: - **Respiratory system** (UBERON:0001004) — respiratory distress, apnea, vocal cord paresis, aspiration. - **Cardiovascular system** — arrhythmia, labile blood pressure. - **Gastrointestinal tract** (UBERON:0001555) — reflux, dysmotility, feeding failure. - **Eye / cornea** (UBERON:0000964 / UBERON:0000411) — alacrima with secondary corneal scarring. - **Musculoskeletal system** — contractures, club foot, hypotonia (secondary to afferent/muscle-spindle loss).

Tissue / cell level. Nervous tissue; specific vulnerable populations are **sensory neurons** (CL:0000101), **peripheral sensory neurons / DRG neurons**, **autonomic neurons**, and **Schwann cells** (CL:0002573). Selectively vulnerable subtypes: NT-3-responsive muscle afferents and GDNF-responsive neurons (PMID: 11241383).

Subcellular level. Axon (GO:0030424); microtubule/IF/actin cytoskeleton; ER membrane (GO:0005789); nuclear envelope (GO:0005635) (PMID: 18638474).

Localization / lateralization. Bilateral and generally symmetric peripheral involvement.

8. Temporal Development

Onset. Predominantly **congenital/neonatal** in severe disease (arthrogryposis at birth, neonatal hypotonia and respiratory distress); an allelic milder form is **adult-onset**. The dystonia musculorum mouse shows that pathology is **restricted largely to postnatal development** despite broad developmental expression of neuronal BPAG1 (PMID: 9242412).

Progression. Progressive degeneration. In the mouse model, massive postnatal DRG neuron degeneration culminates in death at ~4 weeks. In humans, the course ranges from intrauterine/neonatal lethal (LCCS12 end) to slowly progressive over decades (mild adult end).

Patterns. No spontaneous remission. The developmental **critical period** in the mouse is early postnatal life — the window when transgenic DST-a2 restoration partially rescues the phenotype (PMID: 24381311) — suggesting an early therapeutic window in humans.

9. Inheritance and Population

Epidemiology. Ultra-rare. Only ~15 HSAN-VI patients (11 *DST* variants) reported worldwide as of 2025 (PMID: 40938507). Precise prevalence/incidence figures are unavailable; Orphanet classifies it among ultra-rare disorders.

Inheritance. Autosomal recessive (HP:0000007). Biallelic (homozygous or compound heterozygous) variants are required.

Penetrance / expressivity. Penetrance appears **complete** for biallelic LoF, but **expressivity is highly variable**, graded by the specific DST-a isoforms disrupted and by residual DST-A3 compensation.

Genetic anticipation. Not applicable (not a repeat-expansion disorder).

Founder effects / consanguinity. No established founder mutation. **Consanguinity** contributes to homozygous cases (multiple reported families are consanguineous, e.g., PMID: 34897952, PMID: 40938507).

Carrier frequency. Not formally established; expected to be very low given rarity.

Demographics. No sex predilection (autosomal). Cases reported across diverse populations (including a Pakistani family). No endemic geographic clustering.

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10. Diagnostics

Genetic testing is the diagnostic mainstay. HSAN-VI has no pathognomonic biochemical marker; diagnosis rests on identifying **biallelic pathogenic DST-a variants**.

- **Whole exome / whole genome sequencing (WES/WGS):** high utility — the typical route to diagnosis after nonspecific presentation. In one case, karyotype, Prader-Willi assay, SMA panel, and myotonic dystrophy panel were all negative before a comprehensive neuropathy panel/sequencing detected the *DST* variant (PMID: 42563873).
- **Hereditary-neuropathy gene panels** including *DST*.
- **Copy-number / structural variant analysis:** important given the very large *DST* locus (single-nucleotide panels may miss deletions/duplications).
- Isoform-aware interpretation is essential — variant position predicts whether DST-a, DST-b, DST-e, or combinations are affected, and hence the disorder (PMID: 40497796).

Supporting clinical/electrophysiologic tests. Nerve conduction studies and sural nerve biopsy show sensory axonal loss; electron microscopy of peripheral nerve can reveal **severe hypomyelination and dramatically reduced fiber density** (PMID: 37431644). Areflexia and absent corneal reflex are clinical signs. Autonomic testing documents dysautonomia (alacrima, labile BP/temperature, arrhythmia).

Differential diagnosis. Other HSAN subtypes (I–V, VII, VIII), familial dysautonomia / Riley–Day syndrome (autonomic overlap), congenital myopathies and arthrogryposis syndromes, spinal muscular atrophy, and myotonic dystrophy (to be excluded, as in the case above).

Screening. No population newborn screening exists. **Cascade carrier testing** of at-risk relatives and **prenatal/preimplantation genetic testing** are available once the familial variants are known.

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11. Outcome / Prognosis

Survival / mortality. Prognosis is **poor in severe forms**. LCCS12-end disease is intrauterine or neonatal lethal; severe HSAN-VI infants often die in early life from respiratory failure, aspiration, or autonomic crises. The dystonia musculorum mouse dies at ~4 weeks. Milder adult-onset forms are compatible with prolonged survival but progressive disability.

Morbidity / function. Very high — profound global developmental delay, no meaningful psychomotor progress in severe cases, dependence on ventilatory and nutritional support, recurrent injury from pain insensitivity (corneal scarring, self-mutilation, painless ulcers/fractures), and autonomic instability.

Complications. Aspiration pneumonia, respiratory failure, corneal ulceration/scarring, cardiac arrhythmia, GI dysmotility/reflux, contractures.

Recovery potential. None with current care; management is supportive only.

Prognostic factors. The strongest predictor is the **molecular genotype** — which DST-a isoforms are ablated and whether DST-A3 is preserved. Variants ablating all three neuronal isoforms (a1/a2/a3) produce the most severe disease; preservation of DST-A3 predicts milder disease (PMID: 29982604, PMID: 40497796).

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12. Treatment

No approved disease-modifying therapy exists. Care is entirely **supportive and multidisciplinary** (NCIT: Supportive Care Intervention):

- **Respiratory:** airway management, CPAP/ventilatory support, treatment of aspiration.
- **Nutrition/GI:** feeding support (NG/gastrostomy), reflux and dysmotility management.
- **Ophthalmologic:** artificial tears/lubrication and corneal protection to prevent scarring from alacrima and absent corneal reflex.

- **Wound/injury prevention:** protective measures against painless ulcers, burns, fractures, and self-injury (analogous to other HSANs).
- **Autonomic:** cardiac rhythm monitoring, blood-pressure and temperature management.
- **Rehabilitation:** physical/occupational therapy for contractures and hypotonia; orthopedic management of club foot/contractures.
- **Genetic counseling** for the family.

Pharmacogenomics / targeted / immunotherapy: none applicable.

Experimental / future therapeutics. The most compelling proof-of-concept is **neuronal DST-a isoform gene restoration**: transgenic re-expression of neuronal dystonin isoform 2 (DST-a2) partially rescued the dt mouse and extended lifespan (PMID: 24381311). This nominates gene-replacement/gene-therapy strategies as a rational (though pre-clinical) direction. No registered clinical trials (NCT) for HSAN-VI–specific therapies are established.

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13. Prevention

Because HSAN-VI is Mendelian with no environmental component, prevention is **reproductive/genetic**, not lifestyle-based.

- **Primary prevention:** genetic counseling for at-risk couples (especially consanguineous families); carrier testing.
- **Secondary prevention:** prenatal diagnosis and preimplantation genetic testing (PGT) once familial variants are identified; cascade testing of relatives.
- **Tertiary prevention:** vigilant supportive care to prevent complications (corneal protection, aspiration precautions, wound/injury prevention, autonomic monitoring).
- **Immunization / public-health / environmental interventions:** not applicable.

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14. Other Species / Natural Disease

Model species / orthologs. The mouse ortholog *Dst* (dystonin/Bpag1; NCBI Gene 13518) is the basis of the naturally occurring and engineered **dystonia musculorum (dt)** mutant. The gene and its cytolinker function are evolutionarily conserved across mammals. No significant naturally occurring companion-animal or wildlife disease equivalent to human HSAN-VI has been prominently reported; the mouse is the dominant natural/genetic model. Evolutionary conservation of the actin–intermediate-filament linker function is demonstrated by the phenotypic parallels between mouse and human (PMID: 24381311).

Zoonotic potential / transmission. None (non-communicable genetic disease).

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15. Model Organisms

Primary model — the dystonia musculorum (dt) mouse. Homozygous *Dst*-mutant mice phenocopy HSAN-VI, exhibiting progressive limb contractures, dystonia, ataxia, dysautonomia, and early postnatal death (~4 weeks) (PMID: 24381311):

> "Phenotypically, dt mice are similar to HSAN-VI patients, manifesting progressive limb contractures, dystonia, dysautonomia and early postnatal death."

Model types available. A rich allelic series exists, disrupting different Dst isoform combinations: - Classical spontaneous alleles (e.g., **Dst-dt-27J**) disrupting A1/A2/A3 → severe disease. - Milder alleles (e.g., **Dst-dt-Tg4**) where **Dst-A3 upregulation** maintains tubulin acetylation and microtubule stability → less severe phenotype (PMID: 29982604):

> "Maintenance of microtubule stability in Dstdt-Tg4 dorsal root ganglia could be attributed to an upregulation in Dst-A3 expression as a compensation for the absence of Dst-A1 and -A2 in Dstdt-Tg4 sensory neurons."

- **Gene-trap alleles** disrupting actin-binding-domain isoforms, with LacZ reporter for expression mapping (PMID: 25195653).
- **Conditional / cell-type-specific** models dissecting contributions: Schwann-cell-specific loss → late-onset neuropathy and sensory ataxia with hypomyelination (PMID: 32445516); afferent-circuit disruption → cardiac arrhythmia (PMID: 42028226); GI-focused analysis → slowed motility, thinned colon mucus, altered microbiota (PMID: 31814231).
- **Rescue model:** transgenic neuronal DST-a2 re-expression partially rescues and extends lifespan (PMID: 24381311).

Phenotype recapitulation. Excellent — the dt mouse reproduces the sensory-autonomic neuropathy, DRG degeneration, cytoskeletal/transport defects, autophagy and ER stress phenotypes, and dysautonomia (cardiac, GI). **Limitations:** motor/dystonic and cerebellar features are more prominent in mouse; species differences in isoform expression and lifespan; the human severe fetal (LCCS12) end is not fully captured by every allele.

Patient-derived in vitro models. Patient fibroblasts show defective actin cytoskeleton organization, delayed adhesion/spreading/migration, and recombinant p.Ala203Glu dystonin that cannot bind actin (PMID: 30371979):

> "Functional studies showed defects in actin cytoskeleton organization and consequent delayed cell adhesion, spreading and migration, while recombinant p.Ala203Glu dystonin loses the ability to bind actin."

Mechanistic Model / Interpretation

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Biallelic LoF variant in neuronal DST-a isoforms (chr6p12.1)
                         │
                         ▼
    Loss of dystonin-a cytolinker in sensory/autonomic neurons
                         │
                         ▼
 Failure to cross-link  actin ── neurofilament ── microtubule
                         │
       ┌─────────────────┼──────────────────┬───────────────────┐
       ▼                 ▼                  ▼                   ▼
 Cytoskeletal       Impaired fast      ER stress /        Impaired
 disorganization    axonal transport   nuclear-envelope   autophagy
 (α-internexin      (ortho + retro)    dysfunction        (↑LC3-II)
  aggregation)
       │                 │                  │                   │
       └─────────────────┴──────────────────┴───────────────────┘
                         ▼
       Microtubule instability (↓tubulin acetylation)
       [Modifier: residual DST-A3 stabilizes microtubules → milder disease]
                         ▼
       Progressive DRG sensory + autonomic neuron degeneration
       (selective loss of NT-3/GDNF-responsive neurons; caspase-3)
                         │
       ┌─────────────────┴───────────────────┐
       ▼                                     ▼
 SENSORY branch                        AUTONOMIC branch
 pain insensitivity, areflexia,        alacrima, labile BP/temp,
 sensory ataxia, painless ulcers       arrhythmia, GI dysmotility,
 (+ muscle-spindle loss → hypotonia)   respiratory/feeding failure
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The unifying interpretation is that HSAN-VI is a **cytoskeletal-linker (spectraplakin) disease of neurons**: dystonin-a is the physical bridge that holds the three neuronal cytoskeletal systems together, and its loss causes structural collapse of the axon with secondary failures of transport, organelle homeostasis, and quality control (autophagy). Sensory and autonomic neurons — with their long axons and high transport demands — are selectively vulnerable. The **isoform-graded severity** (which DST-a isoforms are lost, with DST-A3 as a built-in protective modifier) elegantly explains the clinical spectrum and is the central prognostic and therapeutic insight.

Evidence Base

PMID	Contribution	Type
40938507	Defines OMIM ID, AR inheritance, core phenotype, causal gene, patient count (~15)	Human clinical
40497796	Establishes DST four-disorder allelic series; variant position predicts phenotype	Human clinical + molecular
35276021	Distinguishes DST/BPAG1 loss in HSAN-VI vs EBS; plakin-domain variant ablates all isoforms	Human clinical
37431644	Severe congenital end (arthrogryposis); hypomyelination + reduced fiber density	Human clinical
30371979	Patient-cell actin cytoskeleton defect; p.Ala203Glu loses actin binding	In vitro / human
34897952	Isoform structure (a/b/e); novel homozygous consanguineous variant	Human clinical
24381311	dt mouse phenocopies HSAN-VI; DST-a2 transgene rescue	Model organism
11241383	Selective vulnerability of NT-3/GDNF-responsive neurons	Model organism
29982604	DST-A3 upregulation preserves microtubule stability → milder disease (modifier)	Model organism
12859670	Impaired orthograde + retrograde fast axonal transport	Model organism
26043942	Impaired autophagic flux (↑LC3-II)	Model organism
18638474	ER stress / ER–nuclear-envelope dysfunction in DRG neurons	Model organism
16691115	α-internexin aggregation, caspase-3 activation → apoptosis mechanism	Model organism
8752219	Establishes BPAG1n as actin–IF cytolinker; axonal architecture perturbed in null mice	Model organism
9242412	Postnatal-restricted pathology; muscle-spindle degeneration → ataxia	Model organism
32445516	Schwann-cell-specific Dst loss → late-onset neuropathy, sensory ataxia	Model organism
42028226		Model organism

PMID	Contribution	Type
	Afferent-circuit disruption → cardiac arrhythmia (autonomic mechanism)	
31814231	GI pathology: slowed motility, thinned mucus, altered microbiota	Model organism
25195653	Gene-trap model; actin-binding-domain isoform loss → dt; CNS circuit involvement	Model organism
42563873	Diagnostic odyssey; genetic testing yields diagnosis after negative panels	Human clinical
37603210	Review of tissue-specific DST isoform roles (neural/muscle/skin)	Review
26778567	Overview of neuronal BPAG1 isoform structure/function	Review

Consistency of evidence. The mechanistic model is strongly convergent: multiple independent mouse studies (transport, autophagy, ER stress, cytoskeleton, cell-type-specific and autonomic) and human patient/cell data all point to a single unifying cytolinker-loss mechanism. No study in the reviewed corpus contradicts the core model.

Limitations and Knowledge Gaps

- **Ultra-rarity:** With only ~15 reported patients, human genotype–phenotype data are sparse; frequency estimates for individual phenotypes are qualitative rather than precise percentages, and no formal prevalence/incidence figures exist.
- **Mechanistic inference from mouse:** Much of the causal chain (transport, autophagy, ER stress, autonomic circuits) is demonstrated in the dt mouse and inferred to hold in humans; direct human tissue confirmation is limited.
- **No human natural-history study or validated quality-of-life instrument** is available; prognostic granularity relies on molecular genotype.
- **No epigenetic, biomarker, or metabolomic signature** has been defined for HSAN-VI.
- **Therapeutics are pre-clinical:** gene-restoration rescue is shown only in mouse; no human trials.

- **Isoform-level diagnostics** require careful, position-aware variant interpretation not universally standardized across laboratories.

Proposed Follow-up Experiments / Actions

1. **Establish an international HSAN-VI patient registry** to aggregate genotype–phenotype data, define phenotype frequencies quantitatively, and support natural-history studies — directly addressing the call in PMID: 42563873 for collective analysis.
2. **Systematic isoform-resolution variant curation** in *DST*: build a locus-specific database mapping variant position → affected isoform(s) → predicted disorder (HSAN-VI / EBS3 / CMYO29 / LCCS12) to operationalize the prediction framework from PMID: 40497796.
3. **Advance neuronal DST-a gene-replacement therapy** from the dt-mouse proof-of-concept (PMID: 24381311) toward AAV/gene-editing preclinical programs, testing the early-postnatal critical window.
4. **Test DST-A3 up-modulation** as a therapeutic strategy, given its natural protective effect on microtubule stability (PMID: 29982604).
5. **Human iPSC-derived sensory/autonomic neuron models** from patient variants to confirm the mouse-derived causal chain (transport, autophagy, ER stress) in human cells.
6. **Autonomic phenotyping protocols** (cardiac rhythm, thermoregulation, GI motility) for patients, guided by the mouse circuit findings (PMID: 42028226, PMID: 31814231), to standardize monitoring and complication prevention.
7. **Formalize corneal-protection and injury-prevention care pathways**, given the high morbidity from alacrima, absent corneal reflex, and pain insensitivity.

Report compiled from 5 iterations, 8 confirmed findings, and 25 reviewed papers. Identifiers verified against MONDO/OLS, OMIM, Orphanet, HGNC, Ensembl, UniProt, gnomAD, ClinVar, and HPO/JAX.
